Journal of Transport Geography 94 (2021) 103123

Contents lists available at ScienceDirect

Journal of Transport Geography

journal homepage: www.elsevier.com/locate/jtrangeo

Public transit accessibility approach to understand the equity for public healthcare services: A case study of Greater Mumbai

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ARTICLE INFO

Keywords:
Accessibility
Healthcare services
Equity
Social vulnerability
Public transport
Mumbai

ABSTRACT

Urban public health is one of the most critical yet neglected aspects of urban planning in developing countries such as India. Inequity in access to government healthcare facilities affects the overall urban population and can substantially negatively impact the vulnerable population, who mostly rely on government healthcare services. In this paper, the accessibility measure for healthcare services by public transport is developed using travel time and the number of transit stops (accounting for transit connectivity) for Greater Mumbai. We also identified socially vulnerable wards (administrative units) using a Social Vulnerability Index (SVI), developed based on 16 indicators using Principal Component Analysis (PCA). Developed regression models showed that the proposed accessibility measure explains the coverage and usage of healthcare services better than the traditional accessibility measure, which is based on only aggregate level travel time impedances. South Mumbai has relatively better accessibility for public hospitals and dispensaries, whereas, lower level of accessibility is observed in the eastern part for public healthcare services. Assessment for the spatial inequity based on the Gini index, bivariate Moran's *I*, and mean access value reveals a higher degree of spatial inequity for accessing government hospitals for the slum population. The study developed a decision framework to suggest effective policy measures, which can be prioritised based on SVI to reduce the disparity in the spatial distribution of accessibility to government healthcare systems for vulnerable groups. Our findings can aid transportation and urban planners, health researchers, and policymakers to improve accessibility in under-served areas and give special attention to the needs of the vulnerable population.

1. Introduction

Public healthcare facilities are the crucial components of the overall healthcare systems of any city. Access to healthcare services can be divided into five dimensions: i) affordability (i.e., costs of healthcare services), ii) acceptability (i.e., health service compliance and satisfaction), iii) availability (i.e., adequacy of health service), iv) spatial/geographic accessibility or approachability (i.e., travel impedance between user and service), and v) accommodation (i.e., appropriateness and suitability of health services) (Levesque et al., 2013; Neutens, 2015). Spatial accessibility to urban services plays a vital role in the transportation domain. The concept of accessibility is deep-seated in transport geography and planning. Providing adequate and equitable accessibility has become a priority of urban planners, public health reformers, and policymakers. In South Asian countries, inequality in healthcare access is a critical issue (Rahman et al., 2017). A study conducted in 48 slum areas of Mumbai for 10,754 births highlighted that the preference for private maternity healthcare increases as the socioeconomic status increases (More et al., 2011). The poor and socially vulnerable population are affected not only because of financial burden but also due to loss of wages for the time they spend on the treatment. One of the reasons for high healthcare expenditure is limited access to healthcare in the public sector, which compels patients to seek private healthcare facilities (Mackintosh et al., 2016). The current COVID-19 situation has made the world realise an increased risk of health challenges for vulnerable population (de Souza et al., 2020; Karaye and Horney, 2020). A systematic assessment of public healthcare accessibility can help policymakers use available resources effectively and make appropriate policy decisions. The present study concentrates on the spatial accessibility for primary healthcare units and hospitals in Mumbai, emphasising socially vulnerable populations.

Spatial accessibility received vast attention in the literature, but most studies focused on accessibility to jobs (Boisjoly et al., 2017; Cui et al., 2019; Deboosere and El-Geneidy, 2018; Guzman and Oviedo, 2018).

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https://doi.org/10.1016/j.jtrangeo.2021.103123
Received 5 November 2020; Received in revised form 10 May 2021; Accepted 16 June 2021
Available online 25 June 2021
0966-6923/

Accessibility measures related to healthcare services from the transportation and equity perspective received limited attention (Neutens, 2015), probably because fewer trips are observed to healthcare services than jobs, shopping, and recreational purposes. Measuring spatial accessibility and equity for the healthcare system is of high importance in developing countries, where spatial distribution and coverage problems related to healthcare services are often evident and critical (Neutens, 2015). Many cities in India have grown spatially with a rapid increase in population. However, the city administrations are not able to meet the demand for public healthcare facilities. The healthcare system extensively relies on the spread and the adequacy of the government hospitals and municipal healthcare units. Even though these institutions have different and specific functions, they constitute a continuum of care, both preventive and curative (Bajpai, 2014). While the lower-level healthcare units ought to provide direct preventive and curative services for most common diseases, the higher-level healthcare facilities are supposed to cater specialised healthcare needs.

Accessibility measures the ease by which an individual can reach potential destinations. It is an important measure, which helps planners and policymakers to evaluate the performance of the transportation and land-use system in an urban area. Therefore, an easy and accurate measure of accessibility has become crucial for effective urban transportation planning. Studies related to accessibility for public transportation are important, as planners and policymakers want to enhance transit ridership, build a more sustainable transportation system, and provide affordable movement for the poor. Unlike private mode, the impedance for public transport can include additional factors other than total travel time, improving the accessibility analysis.

The municipal (government) healthcare services in Mumbai consists of i) hospitals, ii) maternity care units, iii) dispensaries, and iv) health posts. In this study, a comprehensive public transit-based accessibility measure is developed accounting for travel time impedance, and the effect of public transit stops in the vicinity of healthcare services. The inequities are assessed using spatial accessibility for different healthcare facilities by public transport mode, a popular mode of travel in Mumbai, accounting for about 70% of the total trips (LEA Associates South Asia Pvt. Ltd., 2008). The study further analysed the degree of inequity for the slum population, which mostly rely on public transport to access healthcare facilities. The Social Vulnerability Index (SVI) is developed at the ward level using 16 socioeconomic indicators, which helps in highlighting the wards with poor accessibility for government healthcare facilities for the socially vulnerable population.

2. Background and literature review

Since the late 1950s, accessibility has been defined in several ways for different purposes. Hansen (1959) defined accessibility as "*the potential of opportunities for interaction.*" As per Ben-Akiva and Lerman (1979), accessibility is "*the benefits provided by a transportation/land-use system.*" Bhat et al. (2000) defined accessibility as "*a measure of the ease of an individual to pursue an activity of a desired type, at a desired location, by a desired mode, and at a desired time.*" Literature suggests that accessibility consists of two major components: i) transportation component–which deals with the ease of travelling, and ii) land use component–which deals with the spatial distribution of facilities. Researchers have highlighted varied applications of GIS in the healthcare arena by mapping the spatial distribution of healthcare needs and utilisation; in monitoring and evaluating the effects of health policy actions; in determining optimal health service locations; in linking the relationships between inequalities in accessibility and health outcomes (Higgs, 2004; McLafferty, 2003). Gage and Calixte (2006) indicated the importance of accessibility in utilising maternal healthcare services in Haiti. Hiscock et al. (2008) observed an inverse relationship between travel time access and utilisation of pharmacies and general practitioners in New Zealand.

There exist many empirical studies on accessibility to healthcare services in the literature. Luo and Wang (2003) used the gravity-based method and floating catchment area (FCA) method to analyse the accessibility to primary healthcare in Chicago. They proposed an enhanced and more comprehensive two-step floating catchment area (E2SFCA) method to measure accessibility to primary care physicians in northern Illinois (Luo and Qi, 2009). Wan et al. (2012) used a three-step floating catchment area (3SFCA) method to minimise the healthcare demand overestimation problem. They concluded that this method helps in providing policymakers with useful healthcare accessibility information. Accessibility analysis to primary healthcare services for indigenous people, using framework synthesis, found that access to healthcare by indigenous patients and their families can be influenced by their cultural and social determinants such as low levels of education and unemployment (Davy et al., 2016). A mixed approach was used to develop a geographical accessibility measure to healthcare services and found that poor conditions of roads were the significant barriers for accessibility to the district hospital in Ghana (Agbenyo et al., 2017). Studies mentioned above mainly focus on private mode based accessibility to healthcare services. But in developing countries, most of the population in urban areas rely on public transport, which can be seen from the fact that Mumbai's suburban rail system is one of the busiest systems in the world (Sahu et al., 2018). Besides, it is well known that most of the underprivileged, elderly, poor population rely on public transport in general. This makes it important to focus on public transit-based accessibility in the healthcare arena. Zhang et al. (2020) used public transit-based accessibility to healthcare services in Shanghai. They identified census tracts in the central part of Shanghai, which has lower public transport-based accessibility to health facilities. In contrast, peripheral parts have adequate accessibility even though there were limited healthcare facilities nearby. Higgs et al. (2017) examined the impact of different modes of travel on associations between healthcare supply and the percentage of elderly patients. They highlighted the importance of public transport, particularly in areas with vulnerable groups such as the elderly, who are more dependent on public transit to access healthcare services.

Many times, accessibility is also commonly used to express the effects of transportation infrastructure development (Gutiérrez et al., 2011; Stepniak and Rosik, 2013; Vickerman et al., 1999). Geurs and Ritsema van Eck (2001) stated that the transport infrastructure component of accessibility could also be characterised by its location and infrastructure supply (Geurs and Ritsema van Eck, 2001). Accessibility analysis using public transport is essential as public transportation systems lead to sustainability and provide affordable mobility for the poor. However, developing public transit accessibility measures is complicated, as various transit-related factors such as travel time, cost, frequency, connectivity, and quality of services can be considered as impedance. But, the traditional accessibility measures have mainly focused on travel time impedance (Boisjoly and El-Geneidy, 2016; Foth et al., 2013; Golub and Martens, 2014; Moreno-Monroy et al., 2018). Recent studies like Kim and Lee (2019) used travel time, alternative routes, and public transit frequency to measure accessibility for job services. Farber et al. (2014) accounted for service frequency in their accessibility measure. Other studies have used additional elements other than travel time in measuring accessibility using public transport (Fayyaz et al., 2017; Owen and Levinson, 2015). Boisjoly and El-Geneidy (2016) presented the temporal variability in public transit-based accessibility. Improvement in public transport infrastructure can have a positive effect on accessibility measure. As the public transport infrastructure around the service can enhance its connectivity with the distant population, it is crucial to consider the effect of transportation infrastructure in the accessibility measurement.

Accessibility is also used as a proxy measure to quantify spatial and social equity from the transportation perspective (Fol and Gallez, 2014). The transportation planning process has started focusing on policies and strategies targeting deprived areas to address social issues (Harzo, 1998; Sanchez, 2008). Due to the geographical distribution of the healthcare system and distance barriers experienced by the rural area population, the research and empirical evidence for access to healthcare facilities have been accrued to rural and sparsely populated areas (Neutens, 2015). In India, access to the healthcare system is relatively better in urban areas than in rural area (Balarajan et al., 2011; Barik and Thorat, 2015). However, the inequity in access to the healthcare system is still a critical issue in cities of developing countries. Many cities in developing countries have sub-standard healthcare availability and need healthcare reforms. In general, the importance of healthcare services cannot be ignored, and the current situations of COVID-19 have highlighted the same, especially in urban areas for the vulnerable population.

Traditional public transit accessibility measure uses travel time impedance. However, recent studies suggest using other transit-related factors in addition to travel time to enhance the representation of public transit accessibility. We use travel time and the number of public transit stops in the proposed accessibility measure. None of the studies has used the number of transit stops in the vicinity to define public transit accessibility, to the best of our knowledge. Further, with limited research on accessibility to the public healthcare system and inequities associated with the vulnerable population, this study offers unique contributions for urban areas in developing countries.

3. Study area and healthcare service data

The study area is Greater Mumbai, the financial capital of India. The population of Greater Mumbai as per the 2011 census is 12.4 million (Office of the Registrar General and Census Commissioner India ORGI, 2011). According to the United Nations, as of 2018, Mumbai is the 7th most populous city globally. The total area of Greater Mumbai is 458.28 sq. km., divided into 577 Traffic analysis zones (TAZ) (with the projected population data of 2018) by the metropolitan planning authority of Mumbai. The city consists of 24 municipal wards, which act as the administration wards. The dataset of government healthcare services are collected from the Municipal Corporation of Greater Mumbai (MCGM) website (Public Health Department - MCGM, 2021). However, the obtained data were not geocoded on a GIS interface. Each healthcare service was geocoded using Google Maps in Greater Mumbai using the name of the healthcare service, area, and pin-code from the database. This process helped create a final dataset of 30 hospitals, 29 maternity care units, 175 dispensaries, and 183 health posts in GIS format for the study area (see Fig. 1). The health post and dispensary are lower-level healthcare units, hospitals and maternity care units are higher-level healthcare facilities. The health post is the smallest healthcare unit, which provides medication for directly observed treatments and basic ailments (cough, cold, fever, gastrointestinal issues, etc.). The dispensary has a doctor for medical check-ups along with some basic laboratory for medical testing. The considered maternity care units only provide maternity care services, whereas hospitals provide most specialised health-related services, which are difficult to handle at health posts and dispensaries.

4. Framework for accessibility measure

Accessibility measure for healthcare services is evaluated based on the gravity and cumulative opportunity model, and its usage can be observed in the existing literature of the health arena (Crooks and

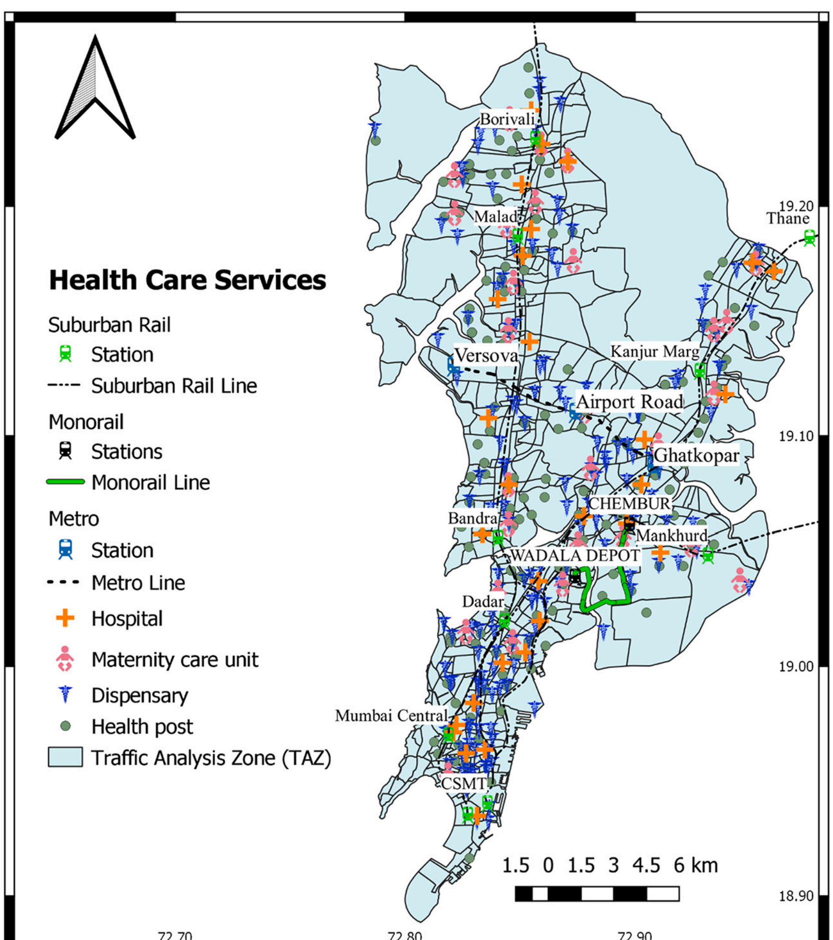

Fig. 1. Spatial distribution of government healthcare services in Greater Mumbai.

Schuurman, 2012; Joseph and Bantock, 1982; Khan, 1992). Hansen (1959) was the first one to put forward the general formulation of the accessibility measure (Acc_i) for a zone i and is given in eq. 1.

$$Acc_i = \sum_{j=1}^{N} S_j d_{ij}^{-x} \tag{1}$$

where, S_j are the opportunities available at zone j, d_{ij} is the travel time or distance between zones i and j, x is an exponent describing the effect of travel time or distance between the zones, and N is the total number of zones considered in the study area.

The current research modified the widely used accessibility measure by incorporating public transportation infrastructure around the service and is discussed in the latter part of this section. We developed the accessibility measure by public transport for government healthcare services, which are likely to be accessed by the low-income population. As mentioned in the literature review, various transit-related factors other than travel time can be used in impedance function in public transit-related accessibility measure. In Mumbai, the public bus stops are closely located at many places, but the buses coming to such bus stops follow different routes. This helps cover more area and enhances the public transit accessibility of that area. Hence, it would be fair to assume that an opportunity or facility appears relatively more attractive if good transportation infrastructure services surround it as compared to the one that lacks. An opportunity or a service surrounded by public transport infrastructure can connect with the distant population over the study area. Thus, we modified the general form of accessibility measure for healthcare services by accounting for the influence of public transportation infrastructure around the service. The transportation infrastructure is defined by the number of transit stops in the vicinity of the service.

The attractiveness of k^{th} unit of service s in zone i is given by $a_{i,k}^s$. The healthcare facilities are grouped into four types: i) hospital, ii) maternity care, iii) dispensary, and iv) health post. The attractiveness value for k^{th} hospital service is the number of beds in that hospital, and it is "1" (number of units) for other healthcare services. A diagrammatic framework is presented in Fig. 2 to explain the methodology for estimating the accessibility measure by considering three zones in a study area. Here the subject zone is zone "*1*", for which the accessibility measure is evaluated by public transport (*PT*) and for service *s* (for simplicity, let us assume one hospital in each zone with different attractiveness). Each service has its attractiveness value, which is represented as a cylinder filled with blue colour mesh pattern given by $a_{i,k}^s$ in Fig. 2. The services in other zones of the study area, i.e., zones (*2* and *3*), also play an essential role in adding a significant proportion of their attractiveness value in estimating accessibility measure for zone *1*. Hence, the accessibility measure (Acc_i^s) of zone i for service s can be written as the sum of two components, as given in eq. 2.

$$Acc_i^s = A_i^s + B_i^s \tag{2}$$

where, A_i^s is the sum of all the attractiveness value of service s in the subject zone i estimated as $\sum_{k=1}^{n_i^s} a_{i,k}^s$ and B_i^s is the proportion of attractiveness values of the other zones in the study area felt in zone i for service s after diminishing effects.

Here, two diminishing effects are considered at different stages. The first diminishing effect is on the attractiveness of the k^{th} unit of service s in other zone j, which is because of public transportation infrastructure near that service s in that zone and is quantified by a factor $\delta_{j,k}^s$ (ranging from 0 to 1). A service s surrounded by good public transportation infrastructure would observe a smaller or no diminishing effect on its attractiveness. As presented in Fig. 2, the value of $\delta_{j,k}^s$ for service in zone *3* is less than the service in zone *2* as the number of bus stops and metro stations are less in number near the service in zone *3*. So, the attractiveness value of the services in other zones j is diminished by multiplying $\delta_{j,k}^s$ (varying from 0 to 1). Then zonal level attractiveness values for service s are estimated by summing all the diminished attractiveness values of all service units of service s in that zone. The second diminishing effect is observed due to travel impedance, which is a function of

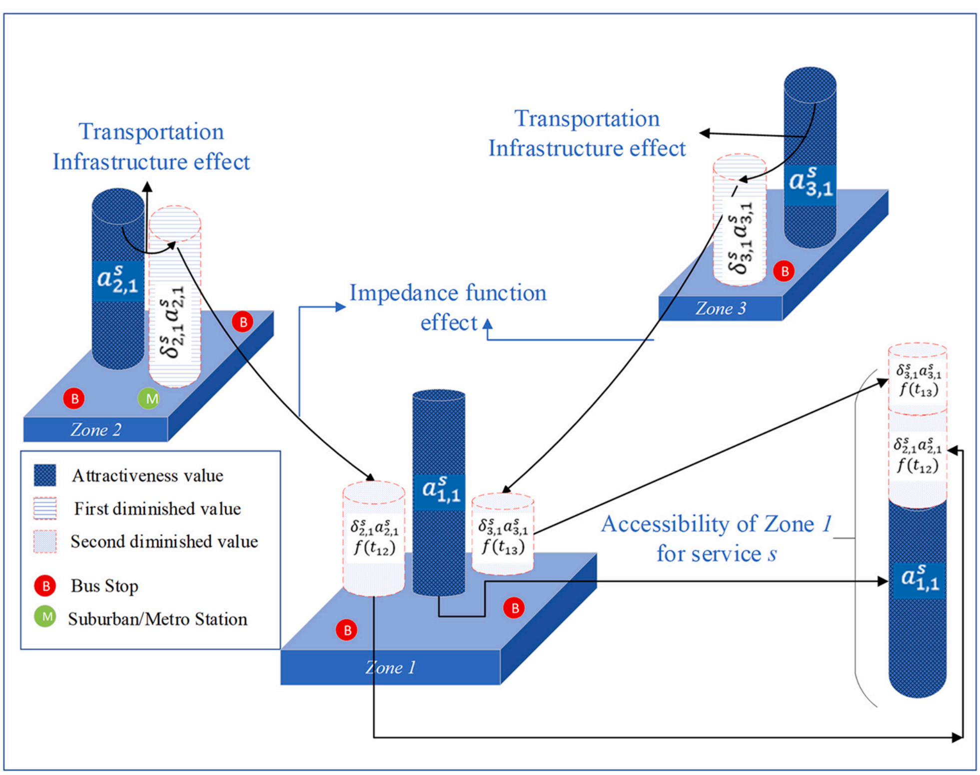

Fig. 2. Diagrammatic framework for accessibility measure of zone 1.

travel cost from zone i to zone j and is given by $f(t_{ij})$. The proportion of attractiveness values felt in zone i because of other zones, i.e., B_i^s can be presented as in eq. 3.

$$B_i^s = \sum_{j=1, j\neq i}^{N} \left(\sum\nolimits_{k=1}^{n_j^s} \delta_{j,k}^s \times a_{j,k}^s \right) f\left(t_{ij}\right) \forall i \text{ and } s \tag{3}$$

The travel cost between zones i and j is considered as travel time between the zones by PT. This resembles the literature that usually calculates accessibility based on travel time (Boisjoly et al., 2017; Van Wee and Geurs, 2011; Cui et al., 2019; Deboosere and El-Geneidy, 2018;). Based on this discussion and understanding related to accessibility, the general formulation for accessibility of a zone i for service s by PT is given in eq. 4.

$$Acc_i^s = \sum_{k=1}^{n_i^s} a_{i,k}^s + \sum_{j=1, j\neq i}^{N} \left(\sum\nolimits_{k=1}^{n_j^s} \delta_{j,k}^s \times a_{j,k}^s \right) f\left(t_{ij}\right) \forall i \text{ and } s \tag{4}$$

where, Acc_i^s= Accessibility of zone i for service s.

$a_{i,k}^s$ = Attractiveness value of k^{th} unit of service s in zone i.

n_i^s = Total number of units of service s in zone i.

$\delta_{j,k}^s$= Public transportation infrastructure parameter near k^{th} unit of service s in zone j ($0 < \delta_{j,k}^s \leq 1$).

$f(t_{ij})$= Impedance function between zone i and j.

t_{ij}=Total travel time from zone i to j by PT.

N = Number of zones in the study area.

The attractiveness value of a service is one of the crucial indicators in understanding the user's potential affinity towards the service. We quantified the attractiveness values based on the healthcare service (see Table 1**;** Attractiveness value component). The improvement in public transport infrastructure can positively affect accessibility measure. The public transportation infrastructure parameter ($\delta_{j,k}^s$) is quantified by considering the number of bus stops and suburban/metro rail stations around the service ($0 < \delta_{j,k}^s \leq 1$). The perception of the individuals are used to quantify public transport infrastructure.

Responses for satisfaction related to public transport was collected from the household survey in Navi Mumbai (an adjacent city in the Mumbai Metropolitan Region). In Mumbai, desired access walking distance changes with respect to the type of transit stop (Rastogi and Rao, 2003). In a review article, it has been found that observed walking distance to bus stops varies from 350 m to 600 m, and for suburban rail stop, it varies from 500 m to 900 m (Vale and Pereira, 2017). Two circular buffers of radius 500 m and 800 m were constructed around respondents' household location to analyse their responses (*Satisfactory = 1* and *Not Satisfactory = 0*). A logistic regression model is developed using the number of bus stops (BS) in a circular buffer of 500 m and the number of suburban/metro rail stations (SRS) in a circular buffer of 800 m. Table 1 presents the results (see Transportation infrastructure parameter component). Using the developed model for Navi Mumbai, $\delta_{j,k}^s$ (public transportation infrastructure parameter) is estimated as the degree of public transport satisfaction based on the number of transit stops near service s by which the attractiveness value of the service s is going to diminish. The model has a positive constant, which suggests that with no transit stops (bus stop and metro/suburban rail stations) in the considered vicinity, the value of $\delta_{j,k}^s$ would be 0.6 and never be zero.

In Fig. 2**,** the second diminishing effect is because of the travel time, i.e., if the travel time between zone i and j increases*,* the diminishing effect reduces the attractiveness of the service. We used two different impedance functions: i) A continuous monotonously decreasing function of travel time between two zones and ii) A weighted dummy function of travel time based on 30 min threshold between two zones. The continuous impedance function used is for home-based other trips consisting of social, recreational, and medical services trips. It was calibrated using the travel data of 2005, collected from the home interview survey by the planning authorities in Mumbai (LEA Associates South Asia Pvt. Ltd., 2008). The calibrated parameters of the continuous impedance function are given in Table 1 (see Impedance function component). The dummy weighted impedance function helps measure the cumulative accessibility, i.e., the number of opportunities reached within a specific threshold travel time. The reason for using 30 min threshold for measuring the cumulative healthcare opportunities is drawn from the literature. Luo and Wang (2003) used a 30 min travel time threshold, based on the recommendation by Lee (1991), to designate shortage areas. Similarly, Gu et al. (2010) used 30 min of travel time threshold to analyse the accessibility to cancer screening clinics in Alberta. The US Department of Health and Human Services also used the 30 min standard criteria to define service areas (Paez et al., 2010).

The zone to zone travel time matrix is extracted from the Bing Maps server (Microsoft Bing Maps, 2019) using API. The total travel time matrix for 577 TAZs is developed by adding the initial waiting time to the extracted public transport travel time from Bing Maps. The zone-to-zone initial waiting time matrix for public transport is used from the Cube Voyager model (Four-step transportation-planning model for Mumbai).

Table 1

Description of different component of the accessibility measure.

Component	Defining attractiveness values of different healthcare units	
	Service	Value of attractiveness ($a_{i,k}^s$)
Attractiveness value ($a_{i,k}^s$)	Hospitals	No. of beds per unit
	Maternity care	1 per unit
	Dispensaries	1 per unit
	Health posts	1 per unit

Component	Transportation infrastructure parameter ($\delta_{j,k}^s$)			
	Variables	Coefficient	SE Coefficient	P-value
Transportation infrastructure parameter ($\delta_{j,k}^s$)	Constant	0.376	0.0794	0.000
	SRS	1.958	0.23	0.001
	BS	0.158	0.00908	0.000
	Model Fit			
	Number of observations		4524	
	Cox and Snell R^2		0.31	
	Value of $\delta_{j,k}^s$ near a k^{th} unit of service s in zone j based on the number of transit stops		$\frac{e^{(0.376+1.958(SRS)+0.158(BS))}}{1+e^{(0.376+1.958(SRS)+0.158(BS))}}$	

Component	Impedance function ($f(t_{ij})$)	
	Different function	Formulation
Impedance function ($f(t_{ij})$)	Continuous function	$(t_{ij})^{\rho}e^{-\beta(lnt_{ij})^2}$; $\rho = 0.001$ and $\beta = 1/3.42$ (LEA Associates South Asia Pvt. Ltd., 2008)
	30 min threshold function	$\begin{cases} 1, & \text{if } t_{ij} \leq 30 \\ 0, & \text{if } t_{ij} > 30 \end{cases}$

5. Spatial distribution of healthcare accessibility

Using the established conceptual framework (Fig. 2), the accessibility measure for each TAZ in the study area for different healthcare services are determined. The spatial distribution of accessibility for different healthcare services by public transport for different impedance functions (30 min threshold and continuous impedance function) is presented in Fig. 3 and Fig. 4. The grouping of accessibility values is based on the *Jenks natural breaks* classification into four different classes. This classification method reduces the variance within the classes and maximises the variance between classes—the darker the green colour, the better the accessibility to healthcare services, and vice-versa. Based on the 30 min threshold accessibility measure, a high level of accessibility is found for

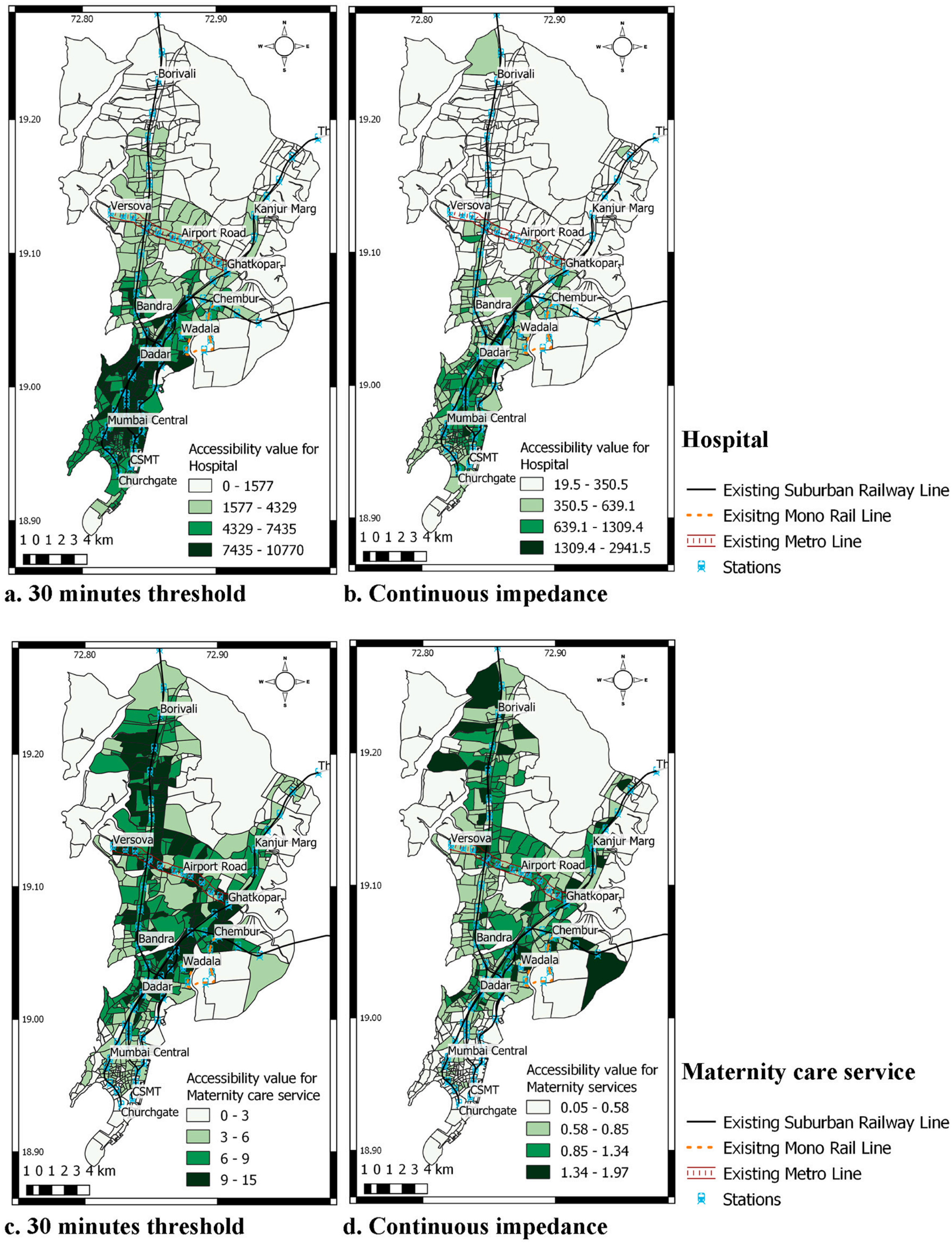

Fig. 3. Spatial distribution of accessibility measures a) for hospitals considering 30 min threshold b) for hospitals considering continuous impedance c) for maternity care services considering 30 min threshold d) for maternity care services considering continuous impedance.

hospitals and dispensaries in the southern part of the study area. In contrast, a relatively high accessibility value is observed for maternity services and health posts in the central part. The spatial distribution of accessibility measure for all the healthcare services tends to be high alongside the mass transit network in the study area (suburban and metro rail network).

Figs. 3 and 4 show that the accessibility measure by continuous impedance function gives lower accessibility values compared to 30 min threshold accessibility measure. This indicates that the continuous impedance function is sensitive to travel time and assign little importance to distant services in the accessibility formulation (see Eq. 4). It can be because of the calibration of the continuous impedance function, in which a high proportion of short trips might be observed. This may depict the travel behaviour for lower-level healthcare units (dispensaries and health posts), which provide direct preventive and curative services for most common diseases.

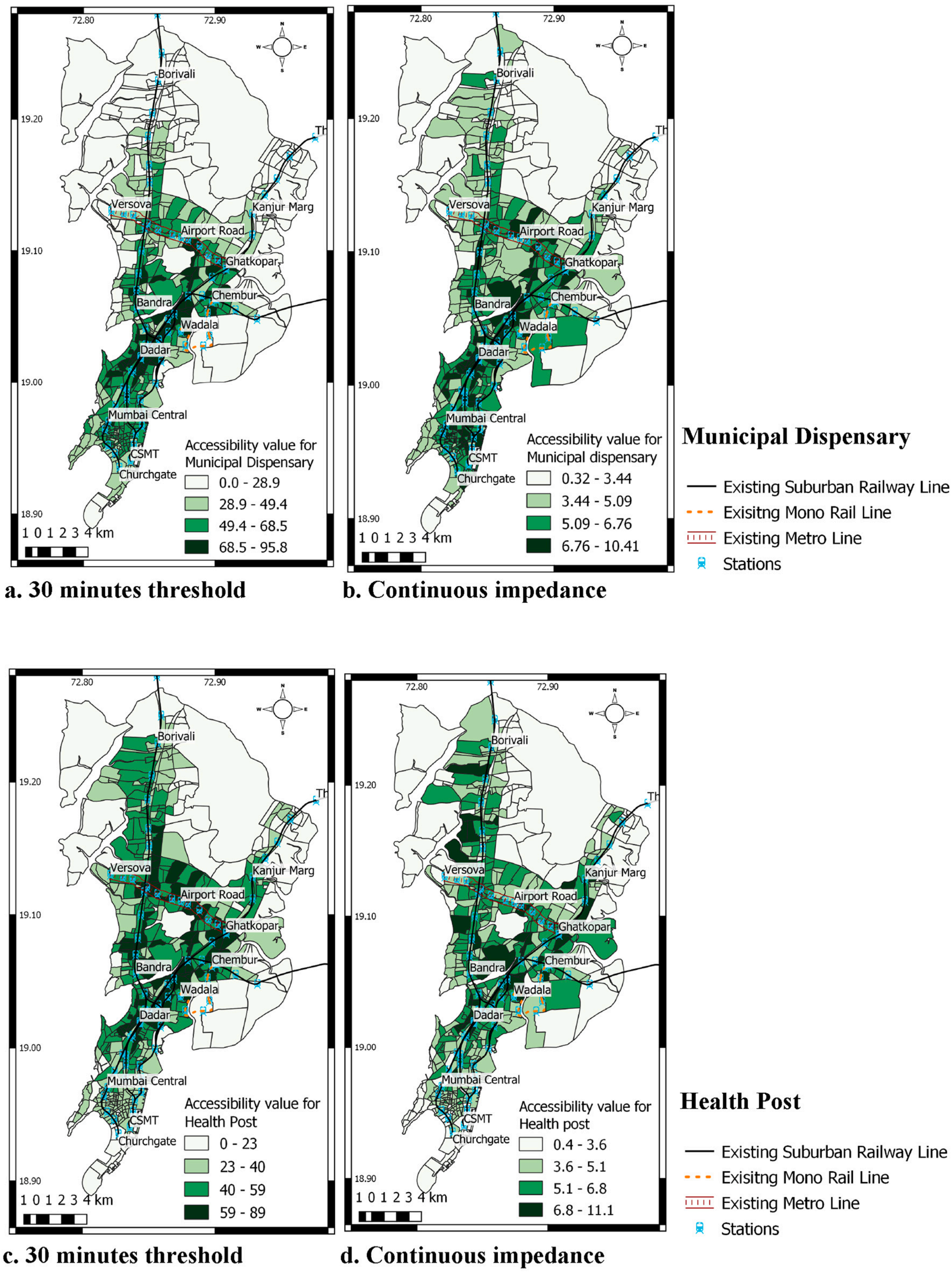

Fig. 4. Spatial distribution of accessibility measures a) for dispensaries considering 30 min threshold b) for dispensaries considering continuous impedance c) for health posts considering 30 min threshold d) for health posts considering continuous impedance.

5.1. Performance of proposed accessibility measure

To measure the performance of our proposed public transit accessibility measure for healthcare services, we examine how well the proposed accessibility measure explains the indicators related to coverage and usage of healthcare services as compared to traditional measures (without considering the connectivity of services with the public transport, i.e., $\delta^{s}_{j,k}$ parameter). We used a linear regression model to link the accessibility measures and healthcare services coverage/usage indicators. Here we assumed that as the public healthcare accessibility for a region increases, the relative coverage/usage of public healthcare services in that region also increases. It can be noted that the indicators related to the public healthcare coverage/usage are available at ward level (bigger spatial units than TAZs), and hence, the traditional and proposed accessibility measures are compared at ward level. The accessibility values at the ward level are estimated as the population-weighted average accessibility values of all TAZs lying in the respective wards. The three indicators to quantify the coverage and usage of public healthcare services are 1) Ward-wise Annual Blood Examination Rate (ABER) in 2017 *(Y_1)*, 2) Ward-wise total OPD patients in municipal dispensaries for August 2018 *(Y_2)*, 3) Ward-wise number of people accessing government hospitals and dispensaries *(Y_3)*. In 2016, India accounted for 90% of the malaria cases in the South East Asia region (World Malaria Report, 2017). There are significant gaps in malaria surveillance, diagnosis, treatment, and control in India (Pradhan et al., 2019; World Malaria Report, 2017). ABER reflects the efficiency and adequacy of the case detection mechanism. Therefore, we used ABER index as a measure of coverage and surveillance for malaria for Mumbai city, where every year malaria outbreak is observed. The ward-wise total OPD patients and the ward-wise number of people accessing public healthcare services represent the usage of public healthcare services. The data of ward-wise ABER in 2017 and ward-wise total OPD patients in municipal dispensaries in August 2018 are extracted from the Public Health Department - MCGM website. The ward-wise number of people accessing government hospitals and municipal dispensaries for 2015 is extracted from the report on State of Health of Mumbai (2015) by ‘Praja’ organisation.

Explanatory variables used in the different models are i) population of the ward (*Pop_ward*), ward-level proposed accessibility values for dispensaries (*Acc_MD*), for hospitals (*Acc_Hos*), and combined for hospitals and dispensaries (*Acc_Hos + MD*), ward-level traditional accessibility values for dispensaries (*Trad_Acc_MD*), for hospitals (*Trad_Acc_Hos*), and combined for hospitals and dispensaries (*Trad_Acc_Hos + MD*). We used accessibility measures related to hospitals and dispensaries in the model because coverage/usage indicators were only available for these services. Table 2 presents the developed regression models between the accessibility measure and the indicators of coverage/usage of public healthcare service. Different models are developed, once using proposed accessibility measure (e.g., Model 1a) and others using traditional accessibility measure (e.g., Model 1b) to compare their performance while modelling each coverage/usage indicators.

Expected signs of the explanatory variables are observed in all the developed regression models. As the population increases, the ABER (surveillance and coverage over the area) decreases. With the increase in population, the total OPD patients in the municipal dispensaries and the number of people accessing government hospitals and dispensaries increases. The sign of variables related to the accessibility measure had a positive association, which suggests that as the accessibility measure related to healthcare services increases, the indicators related to coverage/usage of public healthcare service increases. More importantly, the model goodness of fit, i.e., Adjusted R^2 is better for the models with the proposed accessibility measure as an explanatory variable. For example, Model 1a, which regresses Y_1 (Ward-wise ABER) on *Pop_Ward* and *Acc_Hos* (proposed accessibility value for hospital), has better goodness of fit than Model 1b, where the traditional accessibility value for the hospital is used as an explanatory variable. Similar goodness of fit trends are observed for Y_2 (Ward-wise total patients going to OPD of municipal dispensaries) and Y_3 (Population accessing government hospital and dispensaries) while considering proposed accessibility measure in the models (Model 3a vs 3b and 5a vs 5b). Also, the t-value of the estimate corresponding to the proposed accessibility measure is high compared to the traditional accessibility measure (Model 1a vs1b, Model 3a vs 3b, and Model 5a vs 5b), while linking it to all the three coverage/usage indicators. These results show that the goodness of fit has improved, and the significance of the accessibility variables increased with the use of proposed accessibility measure. Overall, the proposed accessibility measure, which uses both travel time and number of transit stops as impedance, better explains the coverage/usage of public

Table 2
Linear regression models for public healthcare coverage/usage indicators with different accessibility measures.

Response variables	Explanatory variables	Comparing models using continuous impedance for accessibility measure		Comparing models using 30 min threshold impedance for accessibility measure	
		Model 1a	Model 1b	Model 2a	Model 2b
Ward-wise ABER in 2017 (Y_1)	Constant	11.78** (5.22)	13.27** (6.10)	13.28** (6.13	13.28** (6.10)
	Pop_ward	−0.7E-5* (−2.83)	−0.8E-5* (−2.68)	−0.8E-5* (−2.68)	−0.8E-5* (−2.69)
	Acc_ Hos	0.01** (3.56)	–	0.00071** (3.09)	–
	Trad_Acc_ Hos	–	0.0007** (3.01)	–	0.00068** (3.00)
	Adj R^2	0.61	0.56	0.57	0.56
		Model 3a	**Model 3b**	**Model 4a**	**Model 4b**
Total OPD Patients in Municipal Dispensaries in August 2018 (Y_2)	Constant	−9374 (−1.84)	−9245 (−1.49)	−2997 (−0.72)	−2767 (−0.66)
	Pop_ward	0.0287** (6.12)	0.0283** (6.01)	0.0278** (5.66)	0.0274** (5.5)
	Acc_ MD	3065** (3.91)	–	212.7** (3.35)	–
	Trad_Acc_ MD	–	2972** (3.75)	–	204.8** (3.24)
	Adj R^2	0.66	0.64	0.615	0.60
		Model 5a	**Model 5b**	**Model 6a**	**Model 6b**
Number of people accessing government hospital and dispensaries in a ward (Y_3)	Constant	−116171** (−2.93)	−116569** (−2.93)	102201* (−2.82)	−102307* (−2.83)
	Pop_ward	0.564** (12.17)	0.564** (12.20)	0.569** (12.04)	0.571** (12.07)
	Acc_Hos + MD	151.2** (3.04)	–	11.38** (3.00)	–
	Trad_Acc_Hos + MD	–	149** (3.03)	–	11.6** (3.01)
	Adj R^2	0.89	0.87	0.87	0.87

Values in the parenthesis presents *t*-value of the estimates; *: *p*-value < 0.05; **: *p*-value < 0.01.

healthcare services than the traditional accessibility measure.

6. Spatial equity for healthcare services

Equity is a measure of the fair distribution of certain benefits or costs over the population. Lorenz curve and Gini index have been widely used in various fields to understand the distribution of resources over the population, including transportation and especially for accessibility analysis (Ben-Elia and Benenson, 2019; Delbosc and Currie, 2011; Guzman et al., 2017; Jang et al., 2017; Kaplan et al., 2014; Ricciardi et al., 2015; Thomopoulos and Grant-Muller, 2013; Welch and Mishra, 2013; Xia et al., 2016). A general discussion on the Lorenz curve and Gini index are provided in the Appendix. Although there are some limitations in using the Gini index, Sadras and Bongiovanni (2004) highlighted its benefits in observing the spatial variation in terms of inequity. They suggested that distribution of resources/benefits/costs can be considered analogous with the concept of wealth distribution over the population, which is a familiar and relatable concept. Lorenz curves can aid present the distribution of benefits from the transportation system, i.e., accessibility for urban amenities. This can help to quantify inequity from a transportation and urban planning perspective. In this study, the Lorenz curve is used to assess the degree of spatial inequity of accessibility for healthcare services among the different population segments. Here, we have considered three population segments which are: i) total population, ii) slum population, and iii) non-slum population. The projected population of 2018 in each TAZ is used from the Comprehensive Transportation Study (CTS) report (LEA Associates South Asia Pvt. Ltd., 2008), and the slum population in each TAZ is estimated by multiplying the slum population density with the area of the TAZ covered with slums.

In transportation and urban planning, equitable distribution of benefits among the population is of high importance. Inequitable distribution of transportation investments may reinforce social injustice. Fig. 5 represents the Lorenz curve for accessibility measure with 30 min threshold for hospital services. The spatial distribution of accessibility for hospital beds is highly unfair over the study area. It can be noted that access to hospital services for the slum population has a higher degree of spatial inequity (G = 0.886) compared to the total population (G = 0.669).

The Gini index values are derived from the respective Lorenz curve for three different population segments (see Fig. 6) to understand the variability in the degree of spatial inequity for healthcare services based on different impedance functions. This degree of spatial inequity helps in identifying the existence of non-uniform urban development for healthcare services. From Fig. 6, it can be noted, the degree of spatial inequity of accessibility for any type of healthcare service for both the impedance scenarios is higher for the slum population than the non-slum and total population. The highest degree of spatial inequity is observed for 30 min threshold accessibility measure for the hospitals for the slum population (G = 0.884). Although the Gini index is the most relevant summary statistics for the Lorenz curve, it cannot reflect all the information in the data (Weiner and Solbrig, 1984). One of the Gini Index limitations is that similar Gini index values can have different resources distribution. Zhu et al. (2018) pointed that traditional economic methods have limitations in measuring inequity, as they overlook spatial information. Some of the limitations can be handled if additional spatial autocorrelation analysis is performed. In the present context, one major piece of information on the distribution of accessibility for healthcare services for different population segments can be the average accessibility value. Therefore, population-weighted average accessibility values for different healthcare services are estimated for three different population segments. Table 3 presents the average accessibility value for different public healthcare services for different population segments.

The average accessibility value for all the public healthcare services is less for the slum population as compared to the non-slum population. Further, the per cent (%) decrease w.r.t average accessibility value of the total population is higher for public hospitals for the slum population. This difference widens for public hospitals while analysing the accessibility by 30 min threshold impedance. Municipal dispensaries follow a similar trend. Results from Table 3 and Gini index values (see Fig. 6) suggest that the slum population tends to get lower accessibility values for healthcare services with a high variation as far as the distribution over the space is concerned.

To understand the spatial information of accessibility measure for healthcare services for different population segments, we have performed a bivariate Moran's I (I_{xy}) analysis. It is a global measure of spatial autocorrelation, which measures the influence of one variable on the occurrence of another variable in close proximity. The I_{xy} statistic indicates the degree of linear association between one variable, i.e., accessibility to healthcare services (x_i) and a different variable in neighbouring regions, i.e., slum population (y_i). The I_{xy} can be estimated as given in Eq. 5 (Wartenberg, 1985).

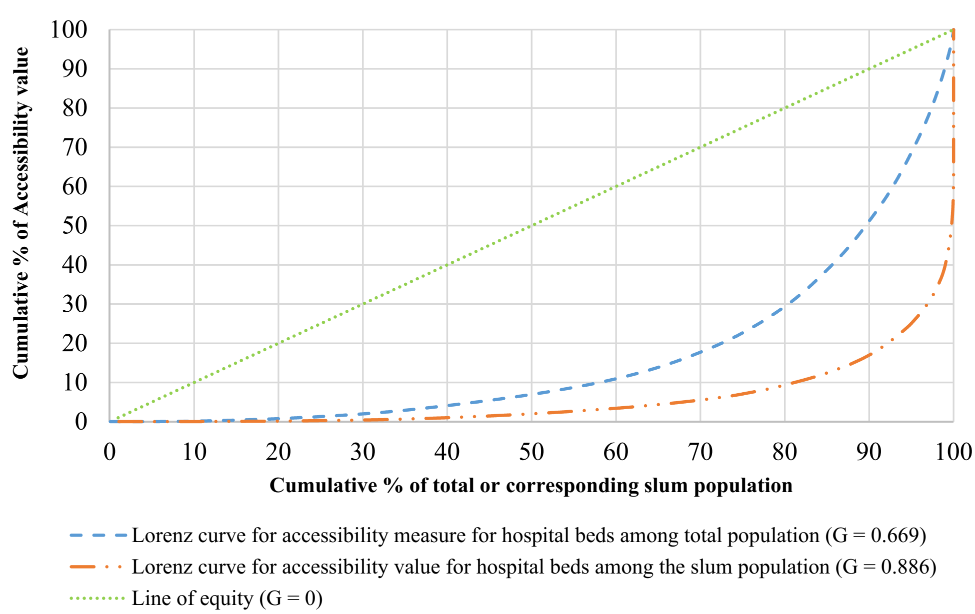

Fig. 5. Lorenz curve for accessibility measure with 30 min threshold for hospital services.

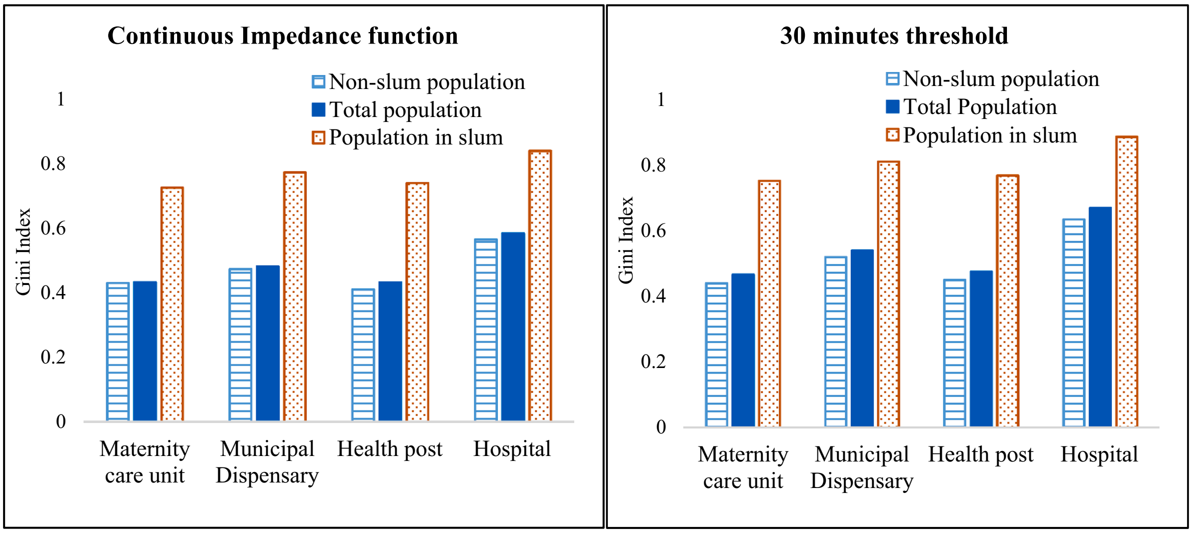

Fig. 6. Degree of spatial inequity of different healthcare accessibility for different impedance functions for different population segments.

Table 3
Average accessibility values for public healthcare services for the different population segments.

	Population segment	Average Accessibility value for healthcare services			
		Municipal Maternity	Municipal Dispensary	Municipal Health post	Public hospital
Accessibility value by continuous impedance	Non-slum population	0.80 (↑1.7%)	4.82 (↑2.3%)	4.75 (↑0.2%)	403.99 (↑10.5%)
	Total population	0.78	4.71	4.74	365.62
	Population in slum	0.75 (↓4.1%)	4.45 (↓5.6%)	4.68 (↓1.2%)	282.72 (↓22.7%)
Accessibility value by 30 min threshold impedance	Non-slum population	6.4 (↑1%)	41.84 (↑5.1%)	39.97 (↑0.8%)	3613.38 (↑13.7%)
	Total Population	6.34	39.82	39.64	3179.33
	Population in slum	6.25 (↓1.4%)	34.39 (↓13.6%)	37.52 (↓5.4%)	2234.19 (↓29.7%)

Values in the parenthesis reflect the % increase (↑) or decrease (↓) in average accessibility values w.r.t total population average values.

$$I_{xy} = \frac{n}{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}} \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}(x_i - \bar{x})(y_j - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_j - \bar{y})^2}} \tag{5}$$

I_{xy} is estimated based on the locational proximity matrices and attribute-similarity matrices. The locational proximity matrix is a spatial weight matrix (W) with elements w_{ij} representing the connectivity between the spatial unit i with the neighbouring spatial unit j. We defined the spatial weight matrix based on the connectivity between the

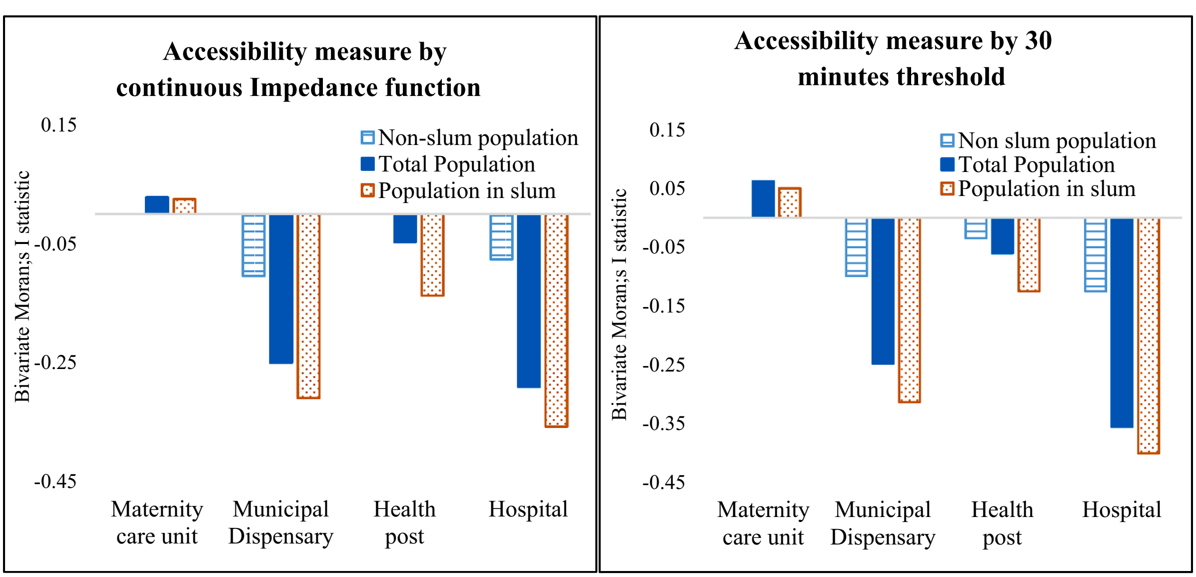

Fig. 7. Bivariate Moran's I (I_{xy}) statistics between the accessibility values for public healthcare services and the population segments in the neighbouring zones.

considered zone i and its neighbouring zone j. The element w_{ij} takes a value of "1" if zone i shares any boundary or corner with zone j and "0" (not connected) otherwise. x_i is one of the attribute of zone i (accessibility to healthcare services), and x^{-} is the mean value of that attribute observed over the space. y_j is the other attribute in neighbouring zone j (population of different segments) and $\overline{y}$ is the mean value of that attribute observed over the space. The value of I_{xy} ranges from -1 to 1. A positive value of I_{xy} indicates that as the value of $(x_i - \overline{x})$ increases the value of $\left(y_j - \overline{y}\right)$ also increases or vice-versa. On the contrary, the value of I_{xy} is negative, when $(x_i - \overline{x})$ increases as $\left(y_j - \overline{y}\right)$ decreases or vice-versa. $I_{xy} = 0$ indicates that the attribute values of the spatial units are arranged randomly and independently in space. Fig. 7 presents the bivariate Moran's I (I_{xy}) value between the accessibility values for public healthcare services and the population segment around it.

The values of I_{xy} statistic presented in Fig. 7 are significant at p-value <0.01. The I_{xy} statistic values are insignificant for accessibility to maternity care units and health post for the non-slum population. It can be seen that the spatial spread of accessibility measure for hospitals, dispensaries, and health posts have a higher (magnitude-wise) negative association with the spatial spread of the slum population in the neighbouring zones compared to the spatial spread of the non-slum population. The highest degree of spatial inequity (Gini index) and the highest negative association (from I_{xy} statistic) over the space is observed for hospitals for the slum population. One possible reason for such high spatial inequity is that most government hospitals are in the southern part of the city, where the slum population is low. It can be observed from I_{xy} statistic that the magnitude of negative association is lowest for municipal health posts and highest for hospitals for the slum population. Similar trends are observed from the Gini index (see Fig. 6). This reveals that healthcare accessibility values are lower in areas surrounded by slum population.

The findings in this section provide most of the information about the spatial distribution of accessibility for healthcare services and provide insights into inequity for healthcare services for different population segments. Higher spatial inequity for all government healthcare services is observed for the slum population compared to the total population, which raises the issue related to social injustice from an urban planning perspective. Therefore, the distribution of public healthcare services needs greater attention for the slum population. They rely primarily on public healthcare services and are at a relatively higher risk of getting affected by diseases.

7. Accessibility for healthcare services and vulnerable population

The use of accessibility measures to address the distributive question, i.e., the distribution of accessibility benefits from investments in the transport system, is crucial to the planners. For this, we suggest that the spatial analysis of accessibility measure can be used for examining and identifying the areas that need closer attention by comparing them with respect to the vulnerable population. For the population living in vulnerable urban communities, government healthcare facilities are essential and often the only source of healthcare (Bhatt and Bathija, 2018).

7.1. Social vulnerability

Social vulnerability refers to the demographic and socioeconomic factors that affect the resilience of communities (Flanagan et al., 2020). Vulnerable populations are at a higher risk of poor health (Aday, 1994; Barata, Ribeiro, Cassanti, and Grupo do Projeto Vulnerabilidade Social no Centro de São Paulo, 2011; Link and Phelan, 1995). With the recent COVID-19 situation, it has been highlighted that socially vulnerable populations are at an increased risk of health challenges (de Souza et al., 2020; Karaye and Horney, 2020). We quantified the social vulnerability at the ward level for Greater Mumbai based on 16 indicators. The indicators reflect four significant aspects: i) Socioeconomic, ii) Demographic, iii) Housing and hygiene, and iv) Poor health status. Five indicators characterise the socioeconomic aspect out of which three reflects asset deprivation as the proportion of households with i) no four-wheeler, ii) no television, and iii) not availing banking services, and other two are iv) population in the slum and v) illiteracy rate. We used five indicators for demographic aspects, which are i) population density, ii) female population, iii) child population, iv) scheduled caste and tribe population, and v) marginal worker population. Housing and hygiene condition are quantified using the proportion of household not connected by i) treated water, ii) electricity, iii) toilets, and iv) drainage. The remaining two indicators, which are i) proportion of underweight students and ii) total deaths, are used to indicate inadequate health status. The indicators used to represent social vulnerability are similar to those proposed by US Centres for Disease Control and Prevention (CDC). The 16 indicators mentioned above and their descriptive analysis are presented in Table A1 (in Appendix).

It is worth noting that the policymaking decisions related to the government healthcare services are mostly taken at the municipal ward level in Mumbai. Hence, for further analysis, it would be fair to focus on the ward level. The data related to the indicators are collected at the municipal ward level (for 24 wards; bigger spatial unit than TAZ) from the Census of India website (Office of the Registrar General and Census Commissioner India ORGI, 2011). All the indicators used in the current study have an increasing effect on social vulnerability, i.e., as the value of the indicator increases, the social vulnerability also increases. We developed a composite score based on 16 indicators using the Principal Component Analysis (PCA) to quantify the Social Vulnerability Index (SVI). More details about the development of the SVI using the PCA technique are provided in the Appendix. Our SVI should be interpreted as a measure of the relative positioning of a ward (or a region) compared with other wards in Greater Mumbai. This relative positioning of the wards can help the planners and policymakers appropriately allocate healthcare resources.

7.2. Mapping healthcare accessibility and socially vulnerable population

In the current section, the accessibility values using 30 min threshold are considered. The accessibility values for different healthcare services are determined at the TAZ level. Since the SVI is developed at the ward level, we estimated accessibility at the ward level. The accessibility values at the ward level are estimated as the population-weighted average accessibility values of all TAZs lying in the respective ward. Ward level accessibility value for different healthcare services and the socially vulnerable groups are compared by mapping spatial cross-classification. The accessibility value and the SVI are divided into two categories. The two categories for each variable are decided based on the average value. The accessibility level for a ward is termed as "Above" if its accessibility value is higher than the mean value of 24 wards. Similarly, the SVI level for a ward is classified as "Below" if the SVI value is less than the average SVI value of 24 wards. A set of four classes based on the below and above categories from two variables are used to cross-classify the spatial units, i.e., wards.

Fig. 8(a-d) presents the spatial cross-classification between accessibility level and SVI level. The four sets of classes based on these level combinations ("Accessibility level-SVI level") are i) Below-Above, ii) Below-Below, iii) Above-Below, and iv) Above-Above. The first level in the class represents the accessibility level, and the second one represents the SVI level. The most critical class would be Below-Above combination, where "Below" represents the low accessibility for healthcare service and "Above" represents a high social vulnerability in the ward. Fig. 8(a-d) shows that the wards in the eastern part of the study area mostly fall in the most critical class. This indicates the disparity in access to healthcare services and highlights the wards for which reasonable healthcare access for the vulnerable population is not met. Further, we

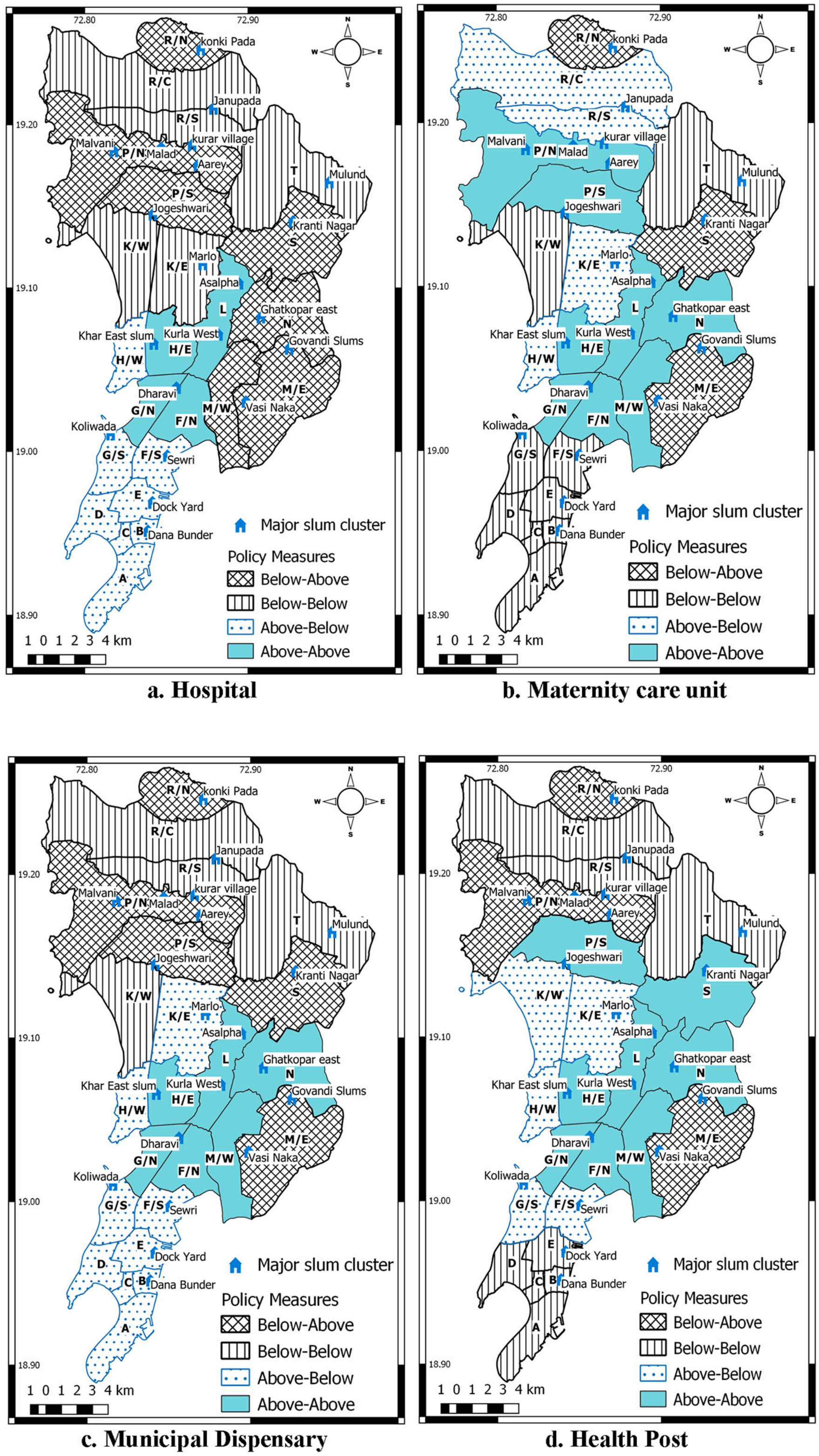

Fig. 8. Spatial cross-classification between accessibility level for different healthcare services and SVI level.

analysed the spatial cross-classification maps to understand the proportion of the population in each of these classes. Table 4 presents the proportion of the population residing in the spatial cross-classification categories.

It can be observed that nearly 35% of the population resides in the most critical combination (Below-Above) from the perspective of accessibility to hospital services. Further, nearly 27% of the population resides in areas where social vulnerability is high, and the available access to municipal dispensaries services is low. A policymaking decision framework is proposed to overcome such a mismatch or gap and enhance the accessibility for public healthcare services. This decision framework considers the current state of the ward, based on the spatial

Table 4
Proportion of population residing in different spatial cross-classification categories.

Spatial cross-classification categories	Percent (%) population residing in spatial cross-classification category			
	Maternity services	Dispensaries	Health post	Hospitals
Below-Above	15.94	27.23	17.52	35.54
Below-Below	24.49	18.84	22.61	25.46
Above-Below	19.17	24.82	21.05	18.20
Above-Above	40.40	29.11	38.82	20.80

cross-classification category.

A decision framework is developed (see Fig. 9) to suggest policy measures from a transportation and urban planning perspective, enhancing public healthcare accessibility. At the first stage, the wards are assessed based on the spatial cross-classification category. As mentioned, the most critical category is Below-Above (with solid red colour outline) classification. But, from the transportation and urban planning perspective, the wards in Below-Below (with dashed orange colour outline) category can also be a concern for policymakers. Hence, wards falling in any of the categories mentioned above are considered, and the decisions can be prioritised based on benefits spread to the maximum population. Once the ward (or region) falls in these critical categories, it is clear that the accessibility to healthcare services is low in that ward. There exist two major policy measures: 1) enhance the public transport system and 2) allocation of public healthcare services in the ward. The decision to improve the public transit system depends on the current state of public transit coverage in the ward. In the suggested framework, public transit coverage is evaluated based on the public bus stop density in the ward. The average public bus stop density in a ward is around 12 bus stops per sq. km. If the bus stop density in a ward is less than 12, then policy measures related to improving the public transit system are suggested. This policy measure can improve the accessibility from a healthcare perspective and enhance the connectivity of the ward for different services and also improve the accessibility for other potential opportunities. But, if a ward has reasonable public transit coverage and low accessibility, then policy measures for public healthcare services expansion are suggested in that ward. Based on the proposed decision framework, Fig. 10 presents the spatial distribution of suggested policy measures. The colour combination used in Fig. 10 for suggesting policy measures are in accordance with the colour combination of the proposed decision framework (Fig. 9).

It can be seen that relatively better healthcare accessibility is observed in the central and the extreme southern part of the study area, and hence, no policy measures are suggested. Overview from Fig. 10 indicates that few areas generally need public transit improvement compared to the expansion (or allocation) of healthcare infrastructure. This indicates that the public transit coverage is currently reasonable in Mumbai (except in some areas). In the eastern part of the study area, policy measure to the expand (allocation) of healthcare services is suggested. The policy measures for allocating healthcare services in wards M/W, N, P/S, R/S, R/N, and S can be prioritised as they have a higher social vulnerability (highlighted with maroon colour in Fig. 10). These wards mainly lack infrastructure related to hospital services. Policymakers can effectively cluster the wards with similar needs and allocate services to attain maximum benefits by minimum allocation. For example, wards S, N, and M/W can be clustered together, and one specialised public hospital can be allocated in ward N so that the population of wards S and M/E can also be served. Similarly, allocating a public hospital on the eastern side of ward P/S would serve both P/S and S wards. The western side of ward R/N can also be considered for the allocation of healthcare services. This can help enhance the healthcare accessibility of peripheral areas of other wards such as R/C and R/S.

The public transport coverage is reasonable in Mumbai, but improvement in public transport coverage is recommended in wards M/E and P/N. This can be prioritised in these wards, as they also have a higher social vulnerability (highlighted with maroon colour mesh pattern in Fig. 10). The improvement in the public transport system in these wards is mainly to expand the public transport network and provide last-mile connectivity from different areas of these wards to the major mass transit system. Policy measures suggested in other wards B, C, D, E, F/S, G/S, K/E, K/W, R/C, R/S, and T should also be considered, as these wards fall in the Below-Above spatial cross-classification

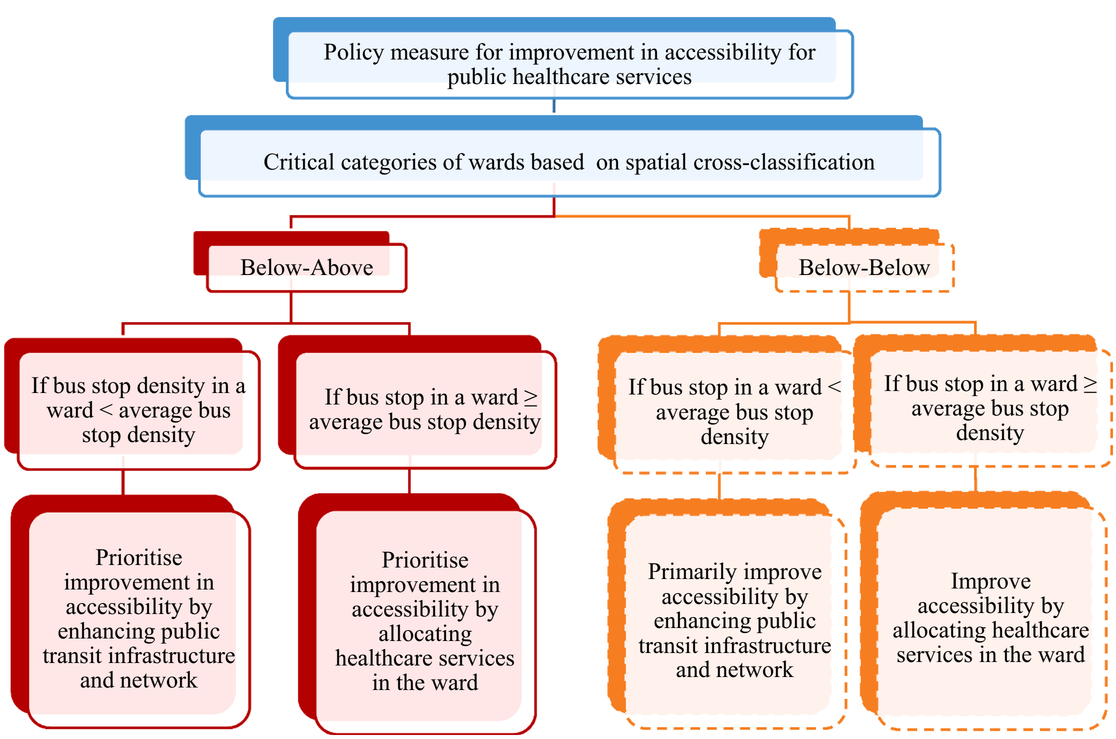

Fig. 9. A decision framework to suggest policy measures to enhance accessibility for healthcare services.

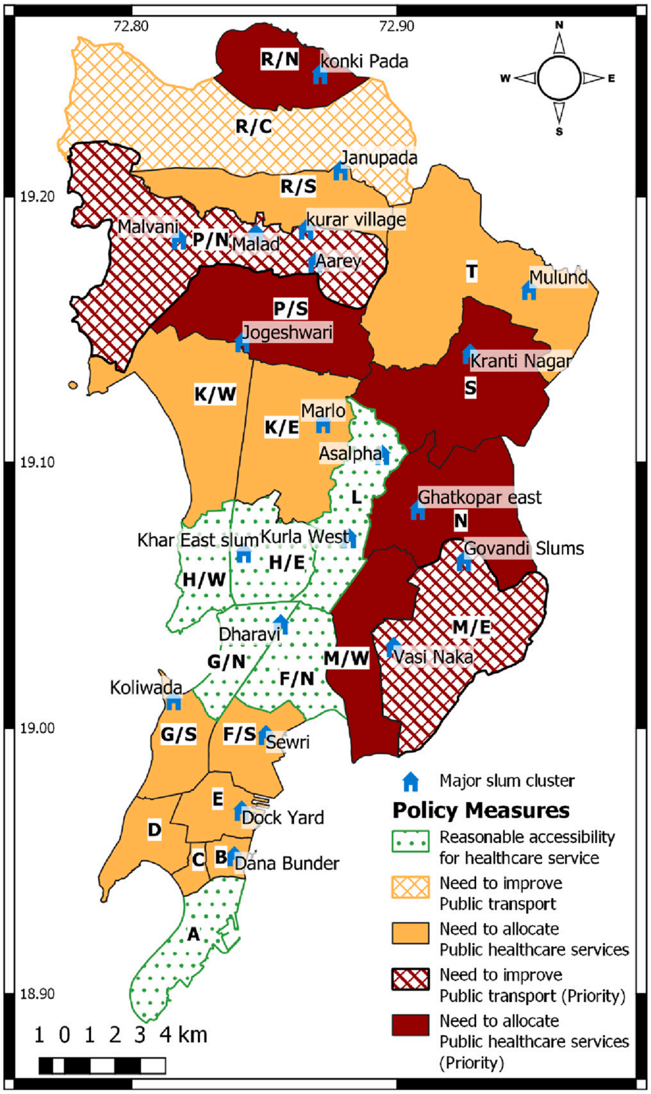

Fig. 10. Policy measures to improve accessibility for healthcare services.

category. Some wards (B, C, D, E, F/S, and G/S) in south Mumbai mainly lack maternity healthcare services. Allocating public maternity services in ward E, surrounded by other wards (B, C, D, F/S, and G/S) would enhance their accessibility. Ward T lacks accessibility for all the public healthcare services; it is recommended to allocate public hospitals, which can also serve ward S. Clustering ward K/E and K/W together and allocation of a hospital in ward K/E, which can also serve ward K/W would enhance their accessibility for the hospital.

8. Conclusions

Accessibility analysis for private mode majorly considers only travel time as the impedance. However, several additional factors like connectivity, frequency, and quality of public transport can affect the accessibility analysis for public transport. Thus, we developed an accessibility measure for public healthcare facilities by public transport that uses travel time impedance and the effect of transit stops. The travel time considered in the proposed accessibility measure consists of in-vehicle travel time and average waiting time. The proposed accessibility measure is evaluated for Greater Mumbai (12.4 million population), which comprises 577 TAZs. Four different types of public healthcare services are considered: hospitals, maternity care units, dispensaries, and health posts. A 30-min threshold and continuous impedance functions are used to estimate the accessibility measure. The spatial inequity for access to public healthcare services is evaluated using the Lorenz curve and Gini index. Additionally, bivariate Moran's *I* and average access value for different population segment are used to understand the degree of spatial inequity. A Social Vulnerability Index (SVI) developed using 16 indicators is used to investigate the spatial cross-classification distribution of accessibility measure for healthcare services among vulnerable urban communities.

South Mumbai has relatively better accessibility for hospitals and dispensary care services. Somewhat better accessibility for maternity and health post is observed in the central part of the study area. The regression models to explain the coverage/usage indicators of public healthcare services indicated that the proposed accessibility measure, which uses both travel time impedance and the effect of transit stops to diminish the attractiveness of the service, performs better than traditional accessibility measures. In general, a higher degree of spatial inequity is observed for hospital services compared to other healthcare services. The average accessibility values for all healthcare services are relatively low for the slum population than the non-slum population. This shows that the spatial inequity for all healthcare services is higher for the slum population, which are highly dependent on such healthcare services.

The spatial cross-classification mapping with respect to accessibility value and social vulnerability suggests that most of the wards in the eastern part of the study area have a relatively high social vulnerability but lack reasonable access to healthcare services. Nearly 35% of the population resides in wards (regions), where the accessibility for hospital services is low and the social vulnerability is high. There is a need to effectively prioritise wards (regions) for either healthcare resource allocation or improvement in the public transportation system to improve the accessibility for healthcare services. Based on the proposed decision framework, wards are classified to identify where and which type of improvements are effective or feasible. This helps in adopting effective strategies for better urban and transportation development from the healthcare perspective. The policymakers and urban/transportation planners can prioritise the suggested measures based on the ward-wise vulnerable population to attain maximum benefits. Also, the transportation and urban planning agencies should analyse public transport-based accessibility. The study findings are beneficial to policymakers and planners to ensure that adequate accessibility for healthcare services is met, especially considering the needs of the disadvantaged population.

9. Limitations and future scope

The estimated accessibility measure of a zone is assumed to be equal for all population residing in that zone. While accessing healthcare services within the zone of residence, no impedance is considered. Most TAZs (zones) have a geographical area of less than one sq. km. Hence, it is reasonable to ignore intra-zonal impedance. In the development of gravity-based accessibility measure, we have used continuous impedance function calibrated for home-based other trip purposes (excluding job, education, and non-home-based trip). The share of these trips was 4.9% and include social, recreational, and medical services trips. The data on explicit medical service trips were not available. While measuring the inequity using the Lorenz curve in some situations, it can be observed that different Lorenz curves can yield similar Gini index values, making it challenging to compare inequity between different population segments. The computation of the Gini coefficient can be sensitive to inequalities in the middle portion of the curve. Using bivariate Moran's *I*, average accessibility value for different population segments, and spatial cross-classification, we overcome some of the limitations of the Gini index.

Average travel time is used to evaluate the accessibility for healthcare services and inequity associated with it, as such measure provides a reasonable understanding of the current scenario. Future studies are encouraged to use the temporal variation of healthcare accessibility, which can help the policymakers and planners understand how the degree of inequities in accessing healthcare services varies at different times of the day.

Acknowledgements

The authors are thankful to two anonymous reviewers for their suggestions, constructive comments, and feedback on previous drafts, which helped improve the manuscript. We would also like to extend our thanks to Mumbai Metropolitan Region Development Authority (MMRDA), Prof. K. V. Krishna Rao at IIT-Bombay and his research group for making us available the zonal level socioeconomic data.

Appendix

A.1. Lorenz curve and Gini index

M. Lorenz, an economist, proposed the Lorenz curve in 1905 to graphically represent the cumulative distribution of wealth among the population (Lorenz, 1905). Gini index is a mathematical value to quantify the overall degree of inequity, which is derived from the Lorenz curve. It is the ratio of the area between the line of equity and the Lorenz curve (marked "A" in Fig. A1) divided by the total area under the line of equity ("A" + "B" in Fig. A1). Gini index (G) ranges from 0 to 1, where close to 0 indicates a fair distribution of resources and close to unity indicates highly unfair distribution.

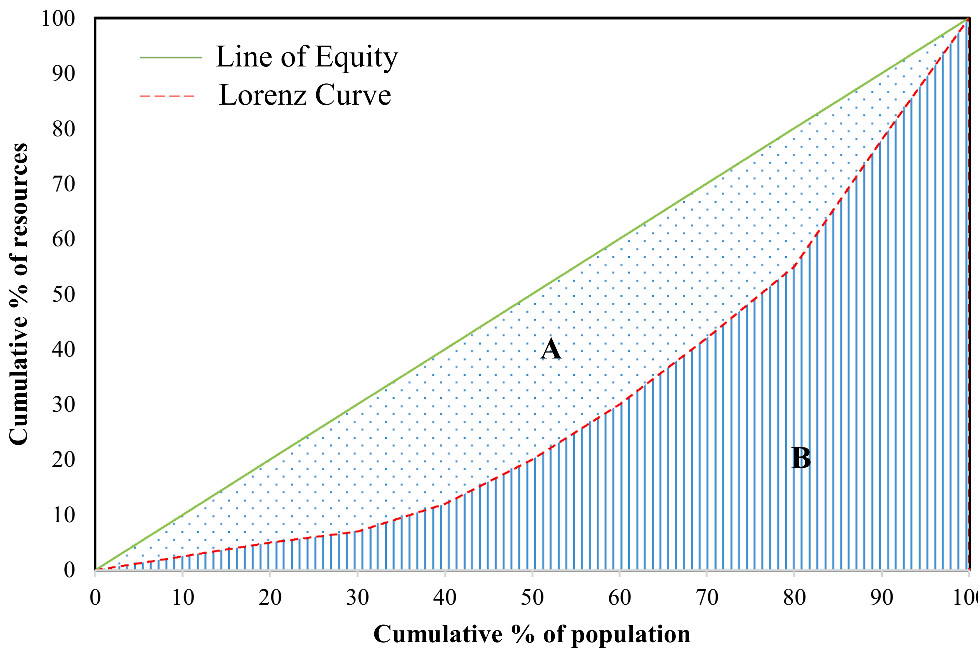

Fig. A1 An example of Lorenz curve.

A.2. Social vulnerability

There is a wide range of application of social vulnerability indices in disaster management and identification of vulnerable people and regions to reduce the ill-effect of natural or human-made hazards on the vulnerable population (Rufat et al., 2015). However, there are no fixed set of factors that determine the social vulnerability of the region or communities. Table A1 presents the descriptive analysis of 16 indicators used to quantify ward level social vulnerability for Greater Mumbai. The indicators reflect four significant aspects, i) socioeconomic, ii) Demographic, iii) Housing and hygiene, and iv) Poor health status.

Table A1 Descriptive statistics of indicators used to develop Social vulnerability index (SVI).

Indicator	Mean	Standard deviation	Minimum	Maximum
Percent (%) of population in slum	44.44	23.39	9.97	84.93
% of household not availing banking service	14.17	8.08	5.76	42.47
% of household not having a television	15.82	7.12	7.67	43.54
% of household not having car/jeep/van	86.89	7.05	70.71	95.18
% of household not availing banking service	14.17	8.08	5.76	42.47
Person illiteracy rate in %	10.15	2.08	6.8	16.6
Population density (persons per sq. km.)	51,286.2	20,740.81	21,072.21	106,401.24
% of female population	45.92	1.54	41	48.56
% of child population (0–6 years)	9.33	1.38	6.75	13.09
% of scheduled castes and tribal population	7.45	3.44	1.6	18.9

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(*continued*)

Indicator	Mean	Standard deviation	Minimum	Maximum
% of marginal workers	2.37	0.44	1.7	3.2
% of household not connected with treated water	2.23	0.96	0.93	4.85
% of household not connected with electricity as the primary source of lighting	3.91	6.88	0.26	35.66
% of household not having latrine facility within the premises	39.16	16.31	6.44	67.89
% of household wastewater outlet not connected with closed drainage	16.79	13.05	1.3	47.9
% of underweight students in municipal schools	5.48	4.48	0	13.4
Total deaths in 2013	3885.58	1575.53	1019	7203

The quantification of the Social Vulnerability Index (SVI) in this study is achieved by developing a composite score based on 16 indicators mentioned in Table A1. The following two assumptions are made while developing SVI: i) Making indicators scale independent and ii) Weights or coefficients for each indicator are estimated using Principal Component Analysis (PCA) technique. It is recommended to transform the indicators as scale-independent or normalise them before using the PCA technique (Christophersen and Hooper, 1992; Jollife and Cadima, 2016). We used Eq a1 to normalise indicator x of v^{th} ward (x_v). In this way, all the normalised indicators range from 0 to 1. For a detailed discussion on PCA, Jollife and Cadima (2016) may be referred.

$$x_v^{'} = \frac{(x_v - Min_x)}{(Max_x - Min_x)} \tag{a1}$$

where, Max_x = maximum value of indicator x, Min_x = minimum value of indicator x, and $x_v = v^{th}$ observation value of indicator x, *and* $x_v^{'}$= normalised value of x_v.

The important aspect while developing such a score or index is assigning weights to the indicators. An equally weighted or simple average of all the indicators tends to make the index very simple. In such a case, we cannot say that one indicator is more important than others in deciding the population's vulnerability. For developing a composite score, numerous methods are followed in the literature. One of the widely used technique is PCA for the development of many indices and for ranking purposes. Berni et al. (2011) used PCA to construct a scale of severity of the vascular risk factors that replicate a continuous ranking. Vyas and Kumaranayake (2006) developed socioeconomic status indices using PCA. PCA has also been used for Internal Market Index, Science and Technology Indicator, Business Climate Indicator (Saisana and Tarantola, 2002) and for the urban quality of life index (Patil and Sharma, 2020). More weight is given to the indicator by using PCA, which is more unequally distributed (Mckenzie, 2003). This would be a fair assumption as an indicator that is good or worst for all the wards (i.e., zero or a small value standard deviation) would exhibit lower variation. Hence, it would be weighted relatively low and so of lesser use in differentiating the SVI. Developing a composite score or index based on weights derived from the first principal component (PC) are also observed in past studies (Houweling et al., 2003; Mckenzie, 2003; Patil and Sharma, 2020). Hence, in the current paper, the coefficients are derived from the first PC of PCA and are used as weights for each indicator, respectively, which can be assumed to be a better approximation than simply averaging the indicators. SVI for a ward v can be estimated by Eq. 2a.

$$SVI_v = \sum_p w_p x_{v,p}^{'} \tag{2a}$$

where $x_{v,p}^{'} = p^{th}$ normalised indicator for ward v and w_p= weight corresponding to indicator p derived from the first PC.

Using the PCA technique, the first PC factors are used as weights to develop SVI (Table A2). The variance explained by the first PC accounts for 41.2% of the total variation. A mean value of 1.38 is observed for the SVI score, and out of 24 wards, 13 wards have an SVI value lower than the mean. It is important to note that as the ward's SVI value increases, the population's vulnerability increases.

Table A2 Estimated weights for different normalised indicators using PCA.

Normalised Indicator	Estimated weights (w_p)
Percent (%) of population in slum	0.294
% of household not availing banking service	0.34
% of household not having a television	0.306
% of household not having car/jeep/van	0.234
Person illiteracy rate in %	0.266
Population density (persons per sq. km.)	0.063
% of female population	0.055
% of child population (0–6 year)	0.339
% of scheduled castes and tribal population	0.148
% of marginal workers	0.157
% of household not connected with treated water	0.266
% of household not connected with electricity as the primary source of lighting	0.238
% of household not having latrine facility within the premises	0.35
% of household wastewater outlet not connected with closed drainage	0.339
% of underweight students in municipal schools	0.191
Total deaths in 2013	0.095

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